Lemma. Let G be a group and A be a set.
For an Action of G on A, $|G : G_a| = |C_a|$

Proof. Since for any $g_1, g_2 \in G_a$, $g_1 g_2^{-1} a = g_1 a = a$, so $g_1 g_2^{-1} \in G_a$.

Thus G_a is a subgroup. And, define a map

$$C_a \to \{gG_a \mid g \in G\} ; g \cdot a \mapsto gG_a$$

Then, $g_1 G_a = g_2 G_a$ implies $g_1^{-1} g_2 \in G_a$, or $g_1^{-1} g_2 a = a$, or $g_1 a = g_2 a$.
So injective, and the surjective is trivial. Thus, Proved.

Thm. Let G be a group and H be a subgroup of G.
An action $G \times \{gH \mid g \in G\} \to \{gH \mid g \in G\} ; (g_1, g_2 H) \mapsto g_1 g_2 H$
satisfies;
1) It acts transitively. (It is clear, since $\{gH \mid g \in G\}$ is a partition of G).
2). $G_H = H$
3) $\ker \pi_H = \bigcap_{x \in G} xHx^{-1}$ is the largest normal subgroup of G contained in H.

Proof. $\ker \pi_H = \{g \in G \mid gxH = xH \text{ for all } x \in G\}$
$= \{g \in G \mid x^{-1} g x H = H \text{ for all } x \in G\} = \bigcap_{x \in G} xHx^{-1}$.
Let $g \in G$ and $h \in \ker \pi_H$. Then, for any $x \in G$,

$$x^{-1}(ghg^{-1})x = (g^{-1}x)^{-1} \cdot h \cdot (g^{-1} \cdot x) \in \ker \pi_H, \text{ so } \ker \pi_H \trianglelefteq G.$$

And, given $g \in \ker \pi_H$ and $h \in H$, $h^{-1} g h H = H \Rightarrow g \in H$, so $\ker \pi_H \leq H$.
Finally, if $N \leq H$ and $N \trianglelefteq G$, then given $x \in G$

$$N = xNx^{-1} \leq xHx^{-1}, \text{ so } N \subseteq \bigcap_{x \in G} xHx^{-1}$$

Corollary. If G is a finite group of order n and p is the smallest prime dividing n, then any subgroup of index p is normal.
Proof. Let $H \leq G$ such that $|G : H| = p$. And, let $K = \ker \pi_H$, π_H is an action above.
Since $K \leq H$, let $|H : K| = k$. Then, $|G : K| = |G : H| \cdot |H : K| = p \cdot k$.
Since H has p left cosets, $G/K \cong \pi_H[G] \leq S_p$, so $pk = |G/K|$ divides $p!$,
so $k = 1$, $H = K$.

「Automorphism of Group.

「Prop. Let H be a normal subgroup of the group G.
Then, G acts by Conjugation on H as automorphisms of H.
More specifically, for each $g \in G$,

$$H \to H : h \mapsto ghg^{-1}$$

is an automorphism of H. Moreover, the permutation represent,

$$\pi_H : G \to S_H : g \mapsto \sigma_g \quad \text{where } \sigma_g(h) = ghg^{-1}$$

is a Homomorphism of G into $Aut(H) \leq S_H$, with $\ker \pi_H = C_G(H)$.
In Summary, $G/\ker \pi_H = G/C_G(H) \cong Im\, \pi_H \leq Aut(H)$, and $G/Z(G) \cong Inn(G)$.

Proof. Since H is normal, the Action is well-defined. And,

$$\ker \pi_H = \{ g \in G \mid \sigma_g = id \} = \{ g \in G \mid ghg^{-1} = h \text{ for all } h \in H \} = C_G(H).$$

「Corollary. For any subgroup H of G,
$N_G(H)/C_G(H)$ is isomorphic to a subgroup of $Aut(H)$.

「Prop. $Aut(\mathbb{Z}_n) \cong \mathbb{Z}_n^\times$.

Proof. Denote $\mathbb{Z}_n = \langle x \rangle$, where $x \in \mathbb{Z}_n$. Given automorphism $\varphi \in Aut(\mathbb{Z}_n)$,
the φ is uniquely determined by $\varphi(x) = x^a$, the integer $1 \leq a \leq n$,
and $|x| = |\varphi(x)| = |x^a|$, so $\gcd(a,n) = 1$. Now,

$$Aut(\mathbb{Z}_n) \to \mathbb{Z}_n^\times ; \varphi_a \mapsto a \bmod n$$

$$\mathbb{Z}_8^\times = \{1,3,5,7\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

is the isomorphism.

Note: If P is a group of a prime order p, then

$$Aut(P) \cong \mathbb{Z}_p^\times \cong \mathbb{Z}_{p-1}.$$

Lemma. Let $P \in Syl_p(G)$. If Q is p-subgroup of G, then $Q \cap N_G(P) = Q \cap P$.

proof. Since $P \subseteq N_G(P)$, $Q \cap P \subseteq Q \cap N_G(P)$ is trivial.

Since $Q \cap N_G(P) \leq N_G(P)$, denote $H = Q \cap N_G(P)$,

$$PH = \bigcup_{h \in H} Ph = \bigcup_{h \in H} hP = HP$$

Thus $PH \leq G$ and $|PH| = \frac{|P| \cdot |H|}{|P \cap H|}$. By the Lagrange's Thm, $H \leq Q$ means order of H is power of p. Therefore, $P \leq PH$ and $|PH|$ is power of p, thus $PH = P$, or $H \leq P$. Now, $H = Q \cap N_G(P) \subseteq Q$, $H \leq Q \cap P$. ⊠

Lemma. If G is finite abelian group and p is a prime number dividing $|G|$, then G has an element of order of p.

Proof Using induction for $|G|$. If $|G| = 1$, trivial.

Now, suppose that the statement hold for all groups whose order is smaller than $|G|$.

Since $|G| > 1$, take $x \in G$ with $x \neq 1$. That is, $|x| > 1$.

If $|G| = p$, then $|x| = p$, by the Lagrange's Theorem $(|x| \mid p)$.

Now, suppose $|G| \neq p$. $(|G| > p)$.

If p divides $|x|$, $|x| = pn$ for some $n \in \mathbb{N}$, so $|x^n| = p$. So clear.

If p does not divides $|x|$, let $N = \langle x \rangle$. (Clearly, $p \nmid |N| = |x|$).

Since G is abelian, N is Normal, and $x \neq 1$, so it is non-trivial.

By the Lagrange Theorem, $|G| = |G/N| \cdot |N|$. But, p does not divide $|N|$, p divides $|G/N|$. Now, using the hypothesis of the induction G/N has an element of order p, denote this $\bar{y} = yN$. Since $\bar{y} \neq \bar{1}$, $y \notin N$.

But, $\bar{y}^p = \bar{1}$, so $y^p \in N$. This implies $\langle y^p \rangle \subsetneq \langle y \rangle$. Since $|y^p|$ divides $|y| = p$ the only possibilities are $|y^p| = 1$, this y is we desired.

「Theorem.

Let G be a group of order $p^\alpha \cdot m$ where p^α is a power of prime p, and $p \nmid m$.

「1st sylow theorem. Sylow p-subgroup of G exists.

「2nd sylow theorem. If $P \in Syl_p(G)$ and Q is p-subgroup of G,

then $Q \leq gPg^{-1}$ for some $g \in G$.

「3rd sylow theorem. $n_p \equiv 1 \mod p$ and $n_p = |G : N_G(P)|$ for any $P \in Syl_p(G)$.

Proof. To show the Existence of Sylow p-subgroup, using induction for $|G|$.
If $|G| = 1$, then $p^\alpha = 1$, so $Syl_1(G) = \{G\} \neq \emptyset$.
Assume that the sylow p-subgroup always exists for any group of order smaller than G.
If order of $Z(G)$ is divided by p, then by the Cauchy's Theorem for abelian group, there exists order of p element $y \in Z(G)$. Let $N = \langle y \rangle$. Denote $\overline{G} = G/N$.
By the hypothesis and the Lagrange's Theorem, $|\overline{G}| = |G|/|N| = p^{\alpha-1} \cdot m$.
By the induction, $\overline{G}$ has a sylow p-subgroup $\overline{P}$ of order $p^{\alpha-1}$.
By the correspondence isomorphism Theorem, there is $P \leq G$ such that $\overline{P} = P/N$.
By the Lagrange's Theorem again, $p^{\alpha-1} = |\overline{P}| = |P/N| = |P|/p$, so $|P| = p^\alpha$.
Therefore, there exists a sylow p-subgroup P, in case of $p \mid |Z(G)|$.
If order of $Z(G)$ is not divided by p, let $g_1, \cdots, g_r$ be representatives of distinct non-central conjugacy classes of G. The class equation gives:

$$|G| = |Z(G)| + \sum_{i=1}^{r} |G : C_G(g_i)|$$

If all $|G : C_G(g_i)|$ are divided by p, then $|Z(G)|$ must be divided by p, Contradiction.
Therefore, for some $1 \leq i \leq r$, p does not divide $|G : C_G(g_i)|$. Let $H = C_G(g_i)$.
By the Lagrange's theorem, $|H|$ divides $|G|$, so $|H| = p^\alpha \cdot k$ where $p \nmid k$, $k \mid m$.
(If $|H|$ does not have factor of p^α, then p divides $|G : C_G(g_i)|$, which is contradiction.)
And, since $g_i \notin Z(G)$, $H < G$. Therefore, by the assumption of induction,
H has a sylow p-subgroup, so it is sylow p-subgroup of G.

To show the second and third assertion, for group G of order $p^{\alpha} \cdot m$, there exists a Sylow p-subgroup P of G, by the first assertion. Let

$$S = \{gPg^{-1} \mid g \in G\} = \{P_1, P_2, \cdots, P_r\}$$

and let Q be any p-subgroup of G.
Let Q acts by conjugation on S. Write S as a disjoint union of orbit:

$$S = \mathcal{O}_1 \cup \cdots \cup \mathcal{O}_s \quad \text{where } |\mathcal{O}_1| + \cdots + |\mathcal{O}_s| = r$$

(Note that S has only one orbit under G, but it may have more than one under Q.)
Renumber the elements of S so that the first s numbers elements of S are representatives of the Q-orbit $\mathcal{O}_i$: $P_i \in \mathcal{O}_i$.

$$\begin{pmatrix} P_1 & P_2 & \cdots & P_s & P_{s+1} & \cdots & P_r \\ \mathcal{O}_1 & \mathcal{O}_2 & \cdots & \mathcal{O}_s & & & \end{pmatrix}$$

It follows that $|\mathcal{O}_i| = |Q : N_Q(P_i)|$.
By definition, $N_Q(P_i) = N_G(P_i) \cap Q$ and
by the Lemma above, $= P_i \cap Q$.
Therefore, $|\mathcal{O}_i| = |Q : P_i \cap Q|$, $1 \le i \le s$. Since Q was arbitrary, take $Q = P_1$. Then $|\mathcal{O}_1| = |Q : Q \cap Q| = 1$. Now, for $2 \le i \le s$, $P_1 \cap P_i < P_1$, so

$$|\mathcal{O}_i| = |P_1 : P_1 \cap P_i| > 1 \quad \text{for all } 2 \le i \le s.$$

Since P_1 has order of power of p, $|\mathcal{O}_i|$ must be power of p. Therefore,

$$r = \underbrace{|\mathcal{O}_1|}_{1} + \underbrace{|\mathcal{O}_2| + \cdots + |\mathcal{O}_s|}_{\text{order of } p} \equiv 1 \mod p. \quad \cdots \quad (*)$$

Let Q be any p-subgroup of G. Suppose that Q is not contained in P_i, for all i.
Then, $Q \cap P_i < Q$ for all $1 \le i \le r$, therefore $|\mathcal{O}_i| = |Q : Q \cap P_i| > 1$ for each $1 \le i \le s$.
Since Q has order of power of p, so $|\mathcal{O}_i| > 1$ is power of p,
therefore all $|\mathcal{O}_i|$ $(1 \le i \le s)$ are divided by p, it contradicts to $(*)$.
Consequently, $Q \le gPg^{-1}$ for some $g \in G$, which proves the second.
Let Q be any Sylow p-subgroup of G. By the argument above, $Q \le gPg^{-1}$ for some $g \in G$. And, since $|Q| = |gPg^{-1}|$, $Q = gPg^{-1}$.
Therefore, all Sylow p-subgroups are conjugate. Finally, for any $P \in Syl_p(G)$,

$$\# Syl_p(G) \overset{\text{def}}{=} n_p = |G : N_G(P)|, \quad \text{since } Syl_p(G) \text{ is the orbit.}$$

Moreover, $m = |G : P| = |G : N_G(P)| \cdot |N_G(P) : P| = n_p \cdot |N_G(P) : P|$, so $n_p \mid m$. ∎

「Corollary.

Let P be a Sylow p-subgroup of G. Then the following are equivalent.

1). P is the unique Sylow p-subgroup of G.

2). P is normal in G

3). P is characteristic in G.

Proof. Start 1). Since $Syl_p(G)=\{P\}$, the Sylow theorem gives that for all $g\in G$, $gPg^{-1}\in Syl_p(G)$, that is, $gPg^{-1}=P$. Therefore $P\trianglelefteq G$.

Conversely, given $Q\in Syl_p(G)$, Q is conjugate of P, that is, $Q=gPg^{-1}$ for some $g\in G$. Since $P\trianglelefteq G$, $Q=gPg^{-1}=P$. ⌟

And, since the automorphism preserves the order, therefore $1)\Rightarrow 3)$.

Finally, $3)\Rightarrow 2)$ trivially, by the inner automorphism, so $3)\Rightarrow 1)$.

「Group of order of 15 is cyclic.
Let G be a group of order 15.
By the Sylow theorem, there are only one Sylow 3-subgroup and Sylow 5-subgroup, denote P, Q respectively. ($n_3 \equiv 1 \mod 3$ and $n_3 \mid 5 \Rightarrow n_3 = 1$)」

Let $x, y \in G$ be generators of P, Q respectively.
Since $P \trianglelefteq G$ is normal, $G = N_G(P)$
Meanwhile, Consider an action of G on P as

$$G \times P \to P : (g, h) \mapsto ghg^{-1}.$$

And the afforded reprsentation $\varphi : G \to S_P$. Then, $\ker \varphi = C_G(P)$, therefore $G/\ker \varphi = G/C_G(P)$ is a subgroup of $\mathrm{Aut}(P)$. But, since $P \cong \mathbb{Z}_3$, $\mathrm{Aut}(P)$ has order of 2, therefore $G/C_G(P)$ must be trivial, that is, $C_G(P) = G$.
Now, x and y commute, so $|xy| = 15$, so $G = \langle xy \rangle$.

To show G must be abelian, suppose that $Z(G) = \{1\}$.
By the Lagrange's theorem, every element of G has order 3 or 5, except 1.
If all non-identity elements have order 3, then the class equation gives

$$15 = 1 + \sum |G : C_G(g_i)| = 1 + k \cdot 5 \text{ for some } k \in \mathbb{N},$$

because $C_G(g_i)$ must contains $\langle g_i \rangle \cong \mathbb{Z}_3$ and if $C_G(g_i) > \langle g_i \rangle$, then $C_G(g_i) = G$ by the Lagrange's theorem, it contradicts to $g_i \notin Z(G)$.
But, $15 - k5$ is divided by 5 but 1 is not divided,
therefore, G must have order 5 element. Put this $x \in G$.

⌈Group of order of pq ($p<q$ are prime, $p \nmid q-1$).

If $Z(G) \neq 1$, then $G/Z(G) \cong \mathbb{Z}_p$ or $\mathbb{Z}_q$ by the Lagrange Theorem, then G must be abelian, since $G/Z(G)$ is cyclic.

Now, suppose that $Z(G)=1$.

By the Lagrange Thm, every non-identity element of G has order p or q.

If there is no order q element, then the class equation gives

$$pq = 1 + \sum |G : C_G(g_i)| = 1 + kq,$$

because $C_G(g_i) = \langle g_i \rangle \cong \mathbb{Z}_p$. (Since $g_i \notin Z(G)$, $C_G(g_i) \neq \langle g_i \rangle$).

This is contradiction, since $q \nmid 1$. So, G has a order q element, denote this to x.

Let $H = \langle x \rangle$. Since $|G:H| = p$ is the smallest prime dividing $|G|$, so H is normal.

Now, $G/H = N_G(H)/C_G(H)$ is a subgroup of order p, and isomorphic to a subgroup of $Aut(H)$.

Since $Aut(H) \cong \mathbb{Z}_q^\times$ has order of $q-1$, this implies $p \mid q-1$, contradiction.

「Group of order of pq, $p<q$. (p, q are prime)
Let $P \in Syl_p(G)$ and $Q \in Syl_p(G)$.
Since n_q divides p, $n_q = 1$ or p, but $n_q \equiv 1 \mod q$, thus $n_q = 1$. $\therefore Q \trianglelefteq G$.
1. Case of $p \nmid q-1$.
By the Sylow Thm, $n_p = 1$ or q, but if $n_p = q$, then $q \equiv 1 \mod p$, $p \mid q-1$, contradiction.
Therefore, $n_p = 1$, $P \trianglelefteq G$. Now, G must be a cyclic, because;
Since $|P|$ and $|Q|$ are primes, both are cyclic. Let $P = \langle x \rangle$ and $Q = \langle y \rangle$.
Since $P \trianglelefteq G$, $N_G(P) = G$. Therefore,

$$G/C_G(P) = N_G(P)/C_G(P) \cong S \text{ for some } S \leq Aut(\mathbb{Z}_p)$$

But, $|Aut(\mathbb{Z}_p)| = p-1$, thus S is trivial. (because $|S|$ divides $|G| = pq$).
Now, $G = C_G(P)$, and $x \in P$, x and y are commute. This implies $|xy| = pq$. $G = \langle pq \rangle$.
2. Case of $p \mid q-1$
Let Q be a Sylow q-subgroup of S_q, the symmetric group q letters.
Since q is prime, $|N_{S_q}(Q)| = q \cdot (q-1)$, because

1). Since $|S_q| = q!$, $|Q| = q$.
2). S_q has $(q-1)!$ q-cycles.
3). Since order of gQg^{-1} is q, these are all cyclic.
4). $gQg^{-1} \cap Q = \{1\}$ for all $g \in G$, so

$$n_q = \#\{gQg^{-1} \mid g \in G\} = (q-1)!/(q-1) = (q-2)!$$

therefore, $n_q = G_Q = [S_q : N_G(Q)] = (q-2)!$, or $N_{S_q}(Q) = q(q-1)$. 」

Now, since $p \mid q-1$, p divides order of $N_{S_q}(Q)$, so it has a subgroup $\overset{P}{\shortparallel}$ of order p.
And, since $P \leq N_{S_q}(Q)$, PQ is subgroup of order pq.
And, $C_{S_q}(Q) = Q$, PQ is non-abelian group.

$$|G : N_{S_q}(Q)| = G_Q$$

「Group of order 48 is ~~Not~~ simple.

$48 = 2^4 \cdot 3$. Therefore,

$$n_3 = 1 \text{ or } 4 \text{ or } 16 \quad \text{and} \quad n_2 = 1 \text{ or } 3.$$

If $n_2 = 1$, then $P \in Syl_2(G)$ is the Normal, so there is nothing to prove.
Suppose $n_2 \neq 1$, or $n_2 = 3$. That is, there are three Sylow 2-subgroup,

$$P_1, P_2, P_3 \subseteq G. \quad (\# P_i = 16).$$

For each $1 \leq i < j \leq 3$,

$$|P_i P_j| = \frac{|P_i| \cdot |P_j|}{|P_i \cap P_j|} = \frac{2^8}{|P_i \cap P_j|} \leq 48 = |G| \quad \text{if } k, P_i = P_j$$

So, $\frac{2^4}{3} \leq |P_i \cap P_j|$, $|P_i \cap P_j|$ cannot be 1, 2, 4, 16 therefore

$$|P_i \cap P_j| = 8.$$

Take $Q = P_1 \cap P_2$, as subgroup of order 8. Since $Q \subseteq P_1, P_2$ and index 2, $Q \trianglelefteq P_1$ and $Q \trianglelefteq P_2$. That is, for any $g \in P_1 \cup P_2$,

$$gQg^{-1} = Q.$$

Therefore, $P_1 \cup P_2 \subseteq N_G(Q)$. Since $|P_1 \cap P_2| = 8$,

$$|P_1 \cup P_2| = |P_1| + |P_2| - |P_1 \cap P_2| = 24$$

Now, if $|N_G(Q)| = 24$, then it has index 2 to G,

$$N_G(Q) \trianglelefteq G$$

And, if $|N_G(Q)| = 48$, then $N_G(Q) = G$, so $Q \trianglelefteq G$.

「Group of order 30 is Not simple.

$30 = 2\cdot3\cdot5$, let $P \in Syl_5(G)$ and $Q \in Syl_3(G)$.

If either P or Q is normal, then $PQ \le G$, and it is group of order 15,

because $|PQ| = \dfrac{|P|\cdot|Q|}{|P\cap Q|} = 15$. (All elements of P has order 5 except identity,
and 〃 Q 〃 3 〃).

So, PQ is index of 2 to G, so PQ is normal subgroup.

Now, assume neither P nor Q is normal. By the Sylow Theorem,
$n_5 = 6$ and $n_3 = 10$. By the Lagrange's Theorem, for any $P_1, P_2 \in Syl_5(G)$,
$P_1 \cap P_2 = \{1\}$. So, there are $6\cdot4 = 24$ distinct order 5 elements.
Similarly, there are 20 distinct order 3 elements, $24+20 = 44 \leq 30$,
Contradiction. Therefore, either P or Q must be normal, so PQ is index 15 subgroup.
In sum, group of order 30 must have a subgroup of order of 15.

「Group of order 105 is Not simple.

$105 = 3\cdot5\cdot7$, so

$$n_3 = 1 \text{ or } 7 \qquad n_5 = 1 \text{ or } 21 \qquad n_7 = 1 \text{ or } 15$$

If it is simple, then $n_3 = 7$, $n_5 = 21$, $n_7 = 15$. Therefore there are

$$\underbrace{2}_{\text{order }3}\cdot7 + \underbrace{4}_{\text{order }5}\cdot21 + \underbrace{6}_{\text{order }7}\cdot15 = 14+84+90 \geq 130$$

So, Contradiction by Counting

⌈Group of order 12 is Not Simple.

Every group of order 12 has a normal Sylow 3-subgroup or $G \cong A_4$.

$12 = 2^2 \cdot 3$. If $n_3 = 1$, then G is not simple, it has a normal sylow 3-subgroup.
Suppose that $n_3 \neq 1$. Then, there is only possibility is $n_3 = 4$.
Let $P \in Syl_3(G)$. By the Sylow Theorem, $|G : N_G(P)| = n_3 = 4$,
so $P = N_G(P)$, by $P \subseteq N_G(P)$ and $|N_G(P)| = 12/4 = 3$, given by the Lagrange Thm.
Now, Consider an action of G on $Syl_3(G)$ by Conjugation:

$$G \times Syl_3(G) \to Syl_3(G) : (g, P) \mapsto gPg^{-1}$$

Since $Syl_3(G) = \{P_1, P_2, P_3, P_4\}$, we can write the permutation representation of this action:

$$\varphi : G \to S_4 : g \mapsto \sigma_g \quad \text{where} \quad \sigma_g : S_4 \to S_4 : P_i \mapsto gP_ig^{-1}$$

Now, the elements of $\ker \varphi$ normalizes all sylow 3-subgroups. That is,

$$\ker \varphi \leq N_G(P) = P$$

Since P is not normal in G, $\ker \varphi = 1$. That is, φ is indective,

$$G \cong \varphi[G] \leq S_4.$$

Since G Contains 8 elements of order 3, by Counting,
and S_4 has exactly 8 element of order 3, and all these are even,
Thus $G \cong A_4$.

「Groups of $p^2 \cdot q$, p and q distinct primes.

Let G be a group of order $p^2 \cdot q$. Let $P \in Syl_p(G)$ and $Q \in Syl_p(G)$.

「In Case of $p > q$, then $n_p = 1$ or $1 + kp$ $(k \geq 1)$, but $1 + kp \nmid q$, so $n_p = 1$.

Thus $P \trianglelefteq G$.

「If $|G|=60$ and G has more than one Sylow 5-subgroup, then G is simple.

Proof. $60 = 5!$.. Since $n_5 > 1$. Then only possibility is $n_5 = 6$.

Assume there is non-trivial proper normal $H \subsetneq G$ exists.

If $5 \mid |H|$, then it has at least one sylow-5 subgroup, and since $H \trianglelefteq G$, all sylow 5-subgroups are contained in H, so,

$$|H| \geq 1 + 4 \cdot 6 = 25,$$

therefore $|H| = 30$. but, since order of 30 has a unique normal sylow 5-subgroup, so contradiction. Thus, $5 \nmid |H|$.

If $|H| = 6$ or 12,